

INTEGRATION SPECIFICATION BROCHURE

EXISTING ERP SYSTEMS & MICROSOFT 365 PLUG & PLAY ENABLEMENT

Executive Operations Summary

This document defines the architectural specifications, integration touchpoints, and utility tools required to deploy Backfeed Agentics. Designed to bridge unstructured commercial inputs with structured enterprise environments, the platform eliminates manual RFQ entry, cross-references regional line cards, detects upsell freight opportunities, and drives warehouse CNC spooling workflows. This specification outlines a 'plug-and-play' methodology to interface with corporate ERP systems and Microsoft 365, ensuring rapid deployment and strict operational continuity.

SYSTEM INTEGRATION NOTICE

This document outlines the standard API, database, and folder-based integration methods for the Backfeed Agentics platform. These specifications are intended to guide IT administrators and integration teams in establishing secure bidirectional data exchange between existing enterprise systems and Microsoft 365.

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Existing ERP Integration Specifications

Backfeed Agentics operates as a bidirectional synchronization layer adjacent to the corporate ERP. By querying customer records, applying pricing contracts, and verifying credit standings in real time, the copilot ensures that incoming distributor inquiries are evaluated against active business rules before routing.

Core ERP Integration Touchpoints

Touchpoint	ERP Data Ingested	Sync Frequency	Operational Impact
Customer Profile	Customer ID, company name, billing address, account standing, credit limit, balance due.	Real-time query per RFQ	Prevents quoting to accounts on hold or without active credit profiles.
Contract Pricing	Customer-specific pricing files, discounts, and represented brand lists.	Batch daily / Real-time query	Ensures exact margin rules are applied to parsed BOM line items.
Credit Evaluation	Order valuation check against available credit limit.	Real-time write check	Automatically triggers credit hold warning if order exceeds credit.
Inventory & Spools	Raw copper/aluminum wire weights, reel capacities, CNC cutting queue.	Real-time query	Optimizes flatbed logistics and machinery cut schedules.
Order Ledger	Finished quote logs, SKU details, quantities, and salesperson allocation.	Real-time write-back	Automatically registers sales volume in ERP ledger.

Integration Methods & Protocols (How to Plug & Play)

- ◆ **REST API (JSON):** Connects via standard HTTPS endpoints using OAuth 2.0. Suitable for modern ERPs like NetSuite, Dynamics 365, or SAP S/4HANA.
- ◆ **Direct DB Connector (SQL/Oracle):** Direct read-only connection to ERP database instances (Prophet 21, Epicor) for high-performance real-time query of inventory.
- ◆ **SFTP Flat-File Exchange:** Automated polling for legacy environments. Exchanges CSV, XML, or XLSX documents on a 15-minute cron cycle.
- ◆ **Custom Agentic Runtime:** Implemented in a dedicated, custom-coded Python/Node.js backend. The architecture explicitly avoids rigid visual flowchart builders (such as n8n or Zapier) to ensure unrestricted licensing for commercial distribution, direct API execution, and robust support for non-deterministic cognitive AI decisions.

Microsoft 365 Integration Specifications

The system interfaces natively with Microsoft 365, turning Outlook and Exchange Online into a centralized sales cockpit. This prevents sales reps from bouncing between email clients, spreadsheets, and web portals.

Core M365 Integration Touchpoints

1. Exchange Online Mail Ingestion (Graph API)

- ◆ Connects via Microsoft Graph API using OAuth 2.0 client credentials flow.
- ◆ Monitors designated sales inboxes (e.g., *quotes@backfeed.com*). Securely extracts raw email bodies, metadata, and attachment bytes (PDF, Excel, CSV) for the BOM Extraction Engine.

2. Outlook Contextual Add-in (Office JS)

- ◆ Built on the modern Office JS framework, compatible with Outlook Desktop, Web (OWA), and Outlook Mobile.
- ◆ Renders a secure sidebar pane directly inside the email interface. When a rep opens a distributor RFQ, the add-in displays the customer's ERP profile, available credit, active brand specialists, and freight upsell progress.

3. SharePoint & OneDrive Document Lifecycle

- ◆ Interfaces with Microsoft Graph Drive API.
- ◆ Automatically archives incoming RFQs in a structured SharePoint folder hierarchy and stores generated PDF proposals, Excel quote sheets, and warehouse dispatch parameters.

4. Microsoft Entra ID (Azure AD) SSO & Security

- ◆ OpenID Connect (OIDC) authentication enforces corporate Single Sign-on (SSO).
- ◆ Inherits security policies, multi-factor authentication (MFA), and role-based access control (RBAC). (e.g., Inside Sales Reps have quote modification rights, while Warehouse Dispatchers only have CNC cutting view permissions).

Graph API Scope Permissions Required

API Scope Name	Permission Type	Integration Purpose
Mail.ReadWrite	Application	Monitors the quote mailbox and marks processed emails with custom categories.
Files.ReadWrite.All	Application	Stores and archives incoming PDF RFQs and outgoing Excel quotes in SharePoint folders.
User.Read	Delegated	Authenticates inside sales representatives and registers their session identities.
Directory.Read.All	Application	Checks Entra ID security group assignments to enforce role-based access control.

Universal Utility Tools & PowerShell Scripts

To maintain operational flexibility and support offline workflows, a suite of universal, brand-agnostic utility tools and PowerShell scripts was developed. These scripts operate independently of third-party SaaS systems, executing directly on local server infrastructure or administrator workstations.

Universal Scripts Inventory

Tool/Script File	Type	Core Functionality	Operational Role
run_ocr.ps1 <i>run_ocr_fixed.ps1</i>	PowerShell	Executes OCR scans on scanned PDF/image documents using Tesseract OCR, outputting clean text.	Document Digitization
convert_pdf.py	Python	Renders PDF pages into high-resolution PNG images for visual extraction validation.	Visual Verification
extract_pdf.py	Python	Programmatically extracts raw text layers from vector PDFs, bypassing complex layout elements.	BOM Text Extraction
inspect_pdf.py	Python	Inspects PDF file health, counting pages, embedded images, and bounding dimensions.	File Quality Control
export_data.py	Python	Compiles line cards and sales representative rosters into CSV, XLSX, and PDF formats.	Master Roster Sync
create_presentation.py	Python	Builds styled widescreen PowerPoint slide decks based on active sales rep and line card datasets.	Presentation Automation
server.py	Python	Runs a local web server to host and test UI dashboards, email simulators, and database views.	Sandbox Testing
take_screenshot.py	Python	Automates headless browser captures of HTML portals and document templates.	Visual Audit Logs

Automation & Scheduling

These tools are designed to require zero SaaS dependencies or licenses, operating entirely on standard Python 3 and PowerShell environments. They can be triggered manually or automated via Windows Task Scheduler, forming a robust utility layer for local data backups and file validation.

Security, Compliance & Deployment Details

Enterprise-grade security policies and data compliance are built directly into the integration framework to ensure data confidentiality, isolation, and protection against modern threat vectors.

Data Security Standards

- ◆ **Data Encryption:** All data is encrypted at rest using AES-256 and in transit using TLS 1.3 encryption, securing all REST and database connections.
- ◆ **Tenant Isolation:** Database schemas enforce tenant-level isolation, preventing cross-customer access and ensuring strict data partitioning.
- ◆ **Domain Threat Verification:** Built-in SPF, DKIM, and DMARC verification processes evaluate domain reputation and registry age metrics to flag potential spoofing or business email compromise (BEC).

Deployment Framework

- ◆ **Cloud SaaS Hosting:** Hosted in a secure, isolated cloud tenant managed by technology partners, utilizing high-availability clusters.
- ◆ **Hybrid Cloud Gateway:** Web services hosted in the cloud, connecting to local ERP databases via a secure IPsec VPN tunnel or private API gateway.

Integration Pre-Flight Checklist

Pre-Flight Requirement	Responsible Team	Status / Core Note
Entra ID App Registration	M365 Admin Team	Required to enable Microsoft Graph API mailbox and file access permissions.
Graph API Permission Consent	M365 Admin Team	Approve Mail.ReadWrite, Files.ReadWrite.All, and Directory.Read.All.
Provision ERP DB Credentials	ERP Admin Team	Create read-only SQL/Oracle user for customer master & pricing database tables.
Configure SFTP / API Gateways	Network Security	Whitelist incoming IP ranges and deploy SSL/TLS certificates.
Deploy Outlook Manifest	M365 Admin Team	Upload manifest XML/JSON file to M365 Admin Center for user sideloading.